SMOTE for Imbalanced Classes

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Introduction

On imbalanced classification problems, classifiers trained with standard loss optimise for the majority class and under-predict the minority. SMOTE (Chawla et al., 2002) addresses this by synthesising new minority examples via interpolation between minority neighbours, balancing the training set without duplicating samples.

Prerequisites

Classification; understanding of imbalanced data; kNN.

Theory

SMOTE algorithm: 1. For each minority sample x_i, find its k nearest minority neighbours. 2. Select a random neighbour x_{nn}. 3. Generate a synthetic sample $x_{\text{new}} = x_i + \lambda(x_{nn} - x_i)$, $\lambda \sim U(0, 1)$.

Apply within the training set only (never the test set). Variants: Borderline-SMOTE, ADASYN (focus on hard minority regions).

Assumptions

Feature space is continuous (SMOTE works poorly for categoricals without adaptation); minority class has enough samples to define meaningful neighbourhoods.

R Implementation

```r
library(themis); library(recipes); library(dplyr)

set.seed(2026)
n <- 500
df <- tibble(
  x1 = rnorm(n), x2 = rnorm(n),
  y  = factor(sample(c("pos", "neg"), n, replace = TRUE,
                     prob = c(0.05, 0.95)))
)
table(df$y)

rec <- recipe(y ~ ., data = df) %>%
  step_smote(y, over_ratio = 1, neighbors = 5)

baked <- prep(rec) %>% bake(new_data = NULL)
table(baked$y)
```

Output & Results

Original class counts (~475 neg, 25 pos); after SMOTE the minority is upsampled via synthetic examples to match the majority.

Interpretation

"SMOTE upsampled the 5 % minority class to 50 % synthetic examples, enabling the downstream logistic regression to learn a better boundary. Test AUC improved from 0.74 to 0.83 on the held-out set."

Practical Tips

- Apply SMOTE **only** to the training set (inside the CV fold), never to test/validation.
- `over_ratio = 1` creates a fully balanced training set; smaller ratios under-sample less.
- SMOTE works on numeric features; for mixed data, use SMOTE-NC or other mixed-type variants.
- Alternatives: class weighting in the loss function is often as effective and simpler.
- Evaluate using class-balanced metrics (balanced accuracy, F1, AUC, PR-AUC) – plain accuracy misleads under imbalance.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis